$$m(x) = \begin{cases} f(x)/|f(x)| & \text{if } f(x) \neq 0 \\ 0 & \text{if } f(x) = 0. \end{cases}$$

Then sign f is the multiplication operator defined by m; i.e.,

$$(\operatorname{sign} f)g = m \cdot g \quad (g \in E),$$

Moreover,

$$(\widehat{\operatorname{sign}} f)g = m \cdot g + \mathbf{1}_{[f(x)=0]}|g| \quad (g \in E).$$

The operator $\widehat{\operatorname{sign}} f$ is related to the Gateaux-derivative (Part-II, Definition 2.2.2) of the modulus. We explain this by an example.

Example 2.2.3 Let E be the real or complex space $L^p(X, \Sigma, \mu)$ where (X, Σ, μ) is a σ-finite measure space and $1 \leqslant p < \infty$. Let $f, g \in E$ and $x \in X$. Then by Part-II, Chapter 2, Lemma 2.4

$$\lim_{t \downarrow 0} 1/t\,(|f(x) + tg(x)| - |f(x)|) = \begin{cases} \operatorname{Re}(\operatorname{sign} \overline{f(x)})g(x) & \text{if } f(x) \neq 0 \\ |g(x)| & \text{if } f(x) = 0. \end{cases}$$

If $\Theta\colon E \to E_+$ denotes the modulus function given by $\Theta(h) = |h|$, then it follows from the dominated convergence theorem that Θ is right-sided Gateaux-differentiable and

$$D_g\Theta(f) = \operatorname{Re}(\widehat{\operatorname{sign}} \bar{f})g. \tag{2.2.6}$$

Later we will see that Eq. (2.2.6) ~~on p. 270~~ holds in every Banach lattice with order continuous norm.

Now let A be the generator of a strongly continuous positive semigroup $(T(t))_{t \geqslant 0}$. The positivity of the semigroup is equivalent to

$$|T(t)f| \leqslant T(t)|f| \quad (t \geqslant 0\,, f \in E). \tag{2.2.7}$$

In order to deduce from Eq. (2.2.7) a property for the generator A it is natural trying to differentiate Eq. (2.2.7) at $t = 0$. Let us assume for a moment that $E = L^p(X, \Sigma, \mu)$ (as in Example 2.2.3). Let $f \in D(A)$ and $0 \leqslant \varphi \in D(A')$. Then by Eq. (2.2.7),

$$\langle |T(t)f|, \varphi\rangle \leqslant \langle T(t)|f|, \varphi\rangle \quad (t \geqslant 0) \tag{2.2.8}$$

where the equality holds for $t = 0$. Hence the inequality remains valid if we differentiate at 0 on both sides of Eq. (2.2.8). Since $\varphi \in D(A')$ we obtain $\mathrm{d}/\mathrm{d}t|_{t=0}\langle T(t)|f|, \varphi\rangle = \langle |f|, A'\varphi\rangle$ on the right side. By Eq. (2.2.6) and the chain rule Part-II, Proposition 2.2.3 one obtains $\mathrm{d}/\mathrm{d}t|_{t=0}|T(t)f| = \operatorname{Re}((\widehat{\operatorname{sign}} \bar{f})Af)$ on the left side. Since $\operatorname{Re}((\operatorname{sign} \bar{f})Af) \leqslant \operatorname{Re}((\widehat{\operatorname{sign}} \bar{f})Af)$, this finally gives

$$\operatorname{Re}\langle (\operatorname{sign} \bar{f})Af, \varphi\rangle \leqslant \langle |f|, A'\varphi\rangle \quad (f \in D(A)\,, 0 \leqslant \varphi \in D(A')). \tag{K}$$

main $C_c^\infty(\mathbb{R}^n)$ generates a strongly continuous positive semigroup on $L^p(\mathbb{R}^n)$ $(1 \leqslant p < \infty)$ (see Example 2.1.5 (iv) and Example 2.4.7).

We want to discuss the relation between the classical (distributional) inequality and our version given in Theorem 2.2.4. Let

$$\mathcal{A} = \sum_{|\alpha| \leqslant m} a_\alpha D^\alpha$$

be a differential operator, where $a_\alpha \in C_c^\infty(\mathbb{R}^n)$. Here we let

$$D^\alpha = (\partial/\partial x_1)^{\alpha_1} \dots (\partial/\partial x_n)^{\alpha_n}$$

for all multi-indices $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$, $(\mathbb{N}_0 := \mathbb{N} \cup \{0\})$ of order $|\alpha| := \alpha_1 + \dots + \alpha_n$. We say that $\mathcal{A}$ satisfies Kato's inequality in the sense of distributions if

$$\operatorname{Re}\langle(\operatorname{sign} \bar{f})\mathcal{A}f, \varphi\rangle \leqslant \langle|f|, \mathcal{A}^*\varphi\rangle \tag{K_d}$$

for all $f \in C_c^\infty(\mathbb{R}^n)$, $0 \leqslant \varphi \in C_c^\infty(\mathbb{R}^n)$, where $\mathcal{A}^*$ denotes the formal adjoint of $\mathcal{A}$.

Let now A be the generator of a positive semigroup $(T(t))_{t \geqslant 0}$ on $E := L^p(\mathbb{R}^n)$ $(1 \leqslant p < \infty)$ or $C_0(\mathbb{R}^n)$. Assume that there exists a core D_0 of A such that $C_c^\infty \subset D_0$ and $Af = \mathcal{A}f$ in the sense of distributions for all $f \in D_0$. Then $\mathcal{A}$ satisfies Kato's inequality in the sense of distributions.

In fact, let $0 \leqslant \varphi \in C_c^\infty(\mathbb{R}^n)$. Then $\langle Af, \varphi\rangle = \langle \mathcal{A}f, \varphi\rangle = \langle f, \mathcal{A}^*\varphi\rangle$ for all $f \in D_0$. Since D_0 is a core of A, this implies that $\varphi \in D(A')$ and $A'\varphi = \mathcal{A}^*\varphi$. Thus ((K) on p. 268) gives $\operatorname{Re}\langle(\operatorname{sign} \bar{f})\mathcal{A}f, \varphi\rangle = \operatorname{Re}\langle(\operatorname{sign} \bar{f})Af), \varphi\rangle \leqslant \langle|f|, A'\varphi\rangle = \langle|f|, \mathcal{A}^*\varphi\rangle$ ~~$= \langle\mathcal{A}|f|, \varphi\rangle$~~ for all $f \in D_0$, $0 \leqslant \varphi \in C_c^\infty(\mathbb{R}^n)$. This is Kato's inequality in the distributional sense.

Remark It has been proved by Miyajima & Okazawa [22]] that ((K_d)) implies that $m \leqslant 2$ and that the principal part $\mathcal{A}_0 = \sum_{|\alpha|=2} a_\alpha D^\alpha$ of $\mathcal{A}$ is elliptic; i.e., if we write the operator $\mathcal{A}_0$ in the form $\mathcal{A}_0 = \sum_{i,j=1}^2 b_{ij}\partial^2/\partial x_i \partial x_j$, then the matrix $(b_{ij}(x))$ is positive semidefinite for all $x \in \mathbb{R}^n$.

Finally we formulate Theorem 2.2.4 for the space $E := C_0(X)$ (X locally compact) (which is not σ-order complete unless X is σ-Stonian). Recall, that for $f \in C_0(X)$, sign f is defined as a Borel function and for any bounded Borel function g on X and any $\varphi \in M(X) = C_0(X)'$, we let $\langle g, \varphi\rangle = \int g(x)\,\mathrm{d}\varphi(x)$ (see Part-II, Section 2.2).

Theorem 2.2.6 *Let X be a locally compact space and A be the generator of a strongly continuous positive semigroup on $C_0(X)$. Then*

$$\operatorname{Re}\langle(\operatorname{sign} \bar{f})Af, \varphi\rangle \leqslant \langle|f|, A'\varphi\rangle \quad (f \in D(A), \varphi \in D(A')_+). \tag{K}$$

2.5 Semigroups of Disjointness Preserving Operators

In this section we consider a special case of domination. Recall from Section 1.6 that a linear operator S on E is called *lattice homomorphism* if

$$|Sf| = S|f| \quad \text{for all } f \in E. \tag{2.5.1}$$

An operator $S \in \mathcal{L}(E)$ is called *disjointness preserving* if

$$f \perp g \quad \text{implies} \quad Sf \perp Sg \quad \text{for all } f, g \in E. \tag{2.5.2}$$

Note that an operator S is a lattice homomorphism if and only if S is positive and disjointness preserving.

In the following we will consider *disjointness preserving semigroups* (by this we mean semigroups of disjointness preserving operators) and *lattice semigroups* (i.e., semigroups of lattice homomorphisms). For example, the semigroup $(T_d(t))_{t\geqslant 0}$ defined in Section 2.3 after Theorem 2.3.13 is disjointness preserving for all $d \in \mathbb{R}$ and a lattice semigroup if $d \geqslant 0$.

Proposition 2.5.1 *A bounded operator S on a complex Banach lattice E is disjointness preserving if and only if there exists a linear operator $|S|$ on E such that*

$$|Sf| = |S||f| \quad (f \in E). \tag{2.5.3}$$

In that case the operator $|S|$ is uniquely determined by Eq. (2.5.3). $|S|$ is a lattice homomorphism and the modulus *of S (i.e., one has $|S| \leqslant T$ for all $T \in \mathcal{L}(E)$ such that $|Sf| \leqslant T|f|$ $(f \in E)$).*

For the proof of the proposition we refer to Arendt [2] and de Pagter [10].

Proposition 2.5.2 *Let $(S(t))_{t\geqslant 0}$ be a disjointness preserving semigroup. Let $T(t) = |S(t)|$ $(t \geqslant 0)$. Then $(T(t))_{t\geqslant 0}$ is a strongly continuous semigroup.*

Proof Let $0 \leqslant s, t$ and $f \in E_+$. Then by Eq. (2.5.1), $T(s)T(t)f = T(s)|S(t)f| = |S(s)S(t)f| = |S(s+t)f| = T(s+t)f$. Since span $E_+ = E$, it follows that $(T(t))_{t\geqslant 0}$ is a semigroup. Moreover, for $f \in E_+$, $\lim_{t\to 0} T(t)f = \lim_{t\to 0} |S(t)f| = |f| = f$. This implies that $(T(t))_{t\geqslant 0}$ is strongly continuous. □

Example 2.5.3 Let $d \in \mathbb{C}$ and $S(t) = T_d(t)$ be given by Eq. (2.3.~~8~~9) on p. 275. Then $|T_d(t)| = T_{|d|}(t)$ $(t \geqslant 0)$.

Proposition 2.5.4 *Let B be the generator of a disjointness preserving semigroup $(S(t))_{t\geqslant 0}$ on a Banach lattice E. Then B is* local*; i.e.*

$$f \perp g \quad \textit{implies} \quad Bf \perp g \quad \textit{for all } f \in D(B), g \in E. \tag{2.5.4}$$

Let $f \in D(B)_+$. Since B is local we have $Bf = P_f Bf = (\operatorname{sign} f)Bf = A|f| = Af$. By assumption, span $D(B)_+ = D(B)$. Thus it follows that $B \subset A$. This implies that $B = A$ since $\varrho(B) \cap \varrho(A) \neq \emptyset$. □

Remark 2.5.12 If B is the generator of a disjointness preserving semigroup $(S(t))_{t\geqslant 0}$ on a real Banach lattice E with order continuous norm then Kato's inequality holds in the reverse sense; i.e.,

$$\langle (\operatorname{sign} f)Bf, \varphi\rangle \geqslant \langle |f|, B'\varphi\rangle \text{ for all } f \in D(B), \varphi \in D(B')_+.$$

(cf. Eq. (2.3.~~9~~10) on p. 275 for a concrete example). In fact, let $T(t) = |S(t)|$ and denote by A the generator of $(T(t))_{t\geqslant 0}$. Let $f \in D(B)$, $\varphi \in D(B')_+$. Then

$$\begin{aligned}\langle (\operatorname{sign} f)Bf, \varphi\rangle &= \langle A|f|, \varphi\rangle \\ &= \lim_{t\to 0}(1/t)\langle T(t)|f| - |f|, \varphi\rangle \\ &\geqslant \lim_{t\to 0} 1/t\langle S(t)|f| - |f|, \varphi\rangle \\ &= \langle |f|, B'\varphi\rangle.\end{aligned}$$

Finally, we come back to Corollary 2.5.9. If in condition (b) we demand that $D(A)$ is not only a sublattice but an ideal of E we obtain a characterization of multiplication semigroups.

Here we call a semigroup $(T(t))_{t\geqslant 0}$ *multiplication semigroup* if $T(t)$ is a multiplication operator (i.e., an element of the center) for every $t > 0$.

Theorem 2.5.13 *Let A be the generator of a strongly continuous semigroup $(T(t))_{t\geqslant 0}$ on a σ-order complete real or complex Banach lattice E. The following assertions are equivalent.*

(a) *$(T(t))_{t\geqslant 0}$ is a multiplication semigroup.*

(b) *There exists $\lambda \in \varrho(A)$ such that $R(\lambda, A)$ is a multiplication operator.*

(c) *$R(\lambda, A)$ is a multiplication operator for all $\lambda \in \varrho(A)$*

(d) *A is local and $D(A)$ is an ideal in E*

(e) *If $f \in D(A)$ then $Pf \in D(A)$ for every band projection P on E and $APf = PAf$.*

Proof Assume that (a) holds and let $\lambda \in \varrho(A)$. Since $R(\lambda, A)$ is the Laplace transform of the semigroup, it follows that $R(\lambda, A)$ is local since $T(t)$ is local for all $t \geqslant 0$. This implies $R(\lambda, A) \in \mathcal{Z}(E)$ (see Section 1.9).

We show that (b) implies (e). Assume that $\lambda \in \varrho(A)$ such that $R(\lambda, A)$ is a multiplication operator. Let P be a band projection. Then $PR(\lambda, A) = R(\lambda, A)P$. Let $f \in D(A)$, $g := (\lambda - A)f$. Then $Pf = PR(\lambda, A)g = R(\lambda, A)Pg$. Hence $Pf \in D(A)$ and $(\lambda - A)Pf = Pg$. Thus $APf = \lambda Pf - Pg = P(\lambda f - g) = PAf$.